



**CALICUT UNIVERSITY – FOUR-YEAR
UNDERGRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	FLUID MECHANICS & THERMODYNAMICS				
Type of Course	Minor (GROUP V: ENERGY PHYSICS)				
Semester	II				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Basic knowledge in units, vectors, pressure, work, mechanical energy and internal energy 2. Basic knowledge about specific heat and molar specific heat capacity				
Course Summary	Students will understand the behavior of fluids, including gas and liquid dynamics, density, pressure, buoyancy, fluid flow, and applications of Bernoulli's equation. Students will also understand the first and second laws of thermodynamics, including entropy, and analyze the directions of thermodynamic processes and will analyze the principles behind heat engines and refrigerators and solve numerical problems based on these topics.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the fluid behavior, the properties of gasses and liquids dynamics including density and pressure in a fluid., buoyancy and fluid flow, applications of Bernoulli's equation.	U	C	Instructor-created exams / Quiz

CO2	Analyze Viscosity and Turbulence in fluids , identifying their effects on fluid behavior.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Grasp the concepts of temperature and thermal equilibrium as well as thermal equilibrium and apply it to calculate the quantity of heat transferred in various processes .	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Understand the first law of thermodynamics and Second law of thermodynamics, and entropy. Analyze the directions of thermodynamic processes and calculate the change in entropy indifferent thermodynamic processes	U	C	Instructor-created exams / Home Assignments
CO5	Analyze the principles behind Heat engines and Refrigerators and solve numerical problems based on these topics.	Ap	P	One Minute Reflection Writing assignments
CO6	Demonstrate comprehension of the second law of thermodynamics, including its application to the Carnot cycle.	Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Fluid Mechanics		10	15
	1	Gasses, liquids and Density, Pressure in a Fluid	2	
	2	Buoyancy, Fluid flow	3	
	3	Bernoulli's Equation	3	
	4	Viscosity and Turbulence	2	
	Sections 12.1, 12.2, 12.3, 12.4, 12.5 and 12.6, Book 1			
II	Temperature and Heat		10	15
	5.	Temperature and Thermal Equilibrium,	1	

	6	thermometers and temperature scales	1	
	7	Thermal Expansion	2	
	8	Quantity of Heat	3	
	9	Mechanisms of Heat Transfer	3	
	Sections 17.1,17.2, 17.3, 17.4, 17.6. Book 1			
III	First Law of Thermodynamics		15	25
	10	Thermodynamic systems	1	
	11	Work done during volume changes	1	
	12	Paths between Thermodynamic states	2	
	13	Internal Energy and First law of Thermodynamics	3	
	14	Kinds of Thermodynamic processes	2	
	15	Internal Energy of an ideal gas	2	
	16	Heat capacities of an ideal gas	1	
	17	Adiabatic process for an ideal gas	3	
	Sections: 19.1, 19.2, 19.3, 19.4, 19.5, 19.6, 19.7, 19.8, Book 1			
IV	The Second law of thermodynamics		10	15
	18	Directions of thermodynamic processes	1	
	19	Heat Engines, Refrigerators	2	
	20	Second law of thermodynamics	2	
	21	The Carnot Cycle	3	
	22	Entropy	2	
	Sections 20.1, 20.2, 20.4, 20.5, 20.6, 20.7, Book 1			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Necessary theory of experiments can be given as Assignment/ Seminar.			
	1	Viscosity of a liquid - Poiseuille's Method		

		● Fill the liquid in a vertically fixed burette with its lower end attached to a capillary tube, placed in horizontal position using a rubber tube. ● Note the time taken to reach each 10cc of water and the height of the corresponding marking. ● Also measure the radius of the capillary tube using the traveling microscope and estimate the viscosity of the liquid.		
2	Viscosity of a liquid - Falling Ball Viscometer	● Drop a polished steel ball into a glass tube of a somewhat larger diameter containing the liquid. ● Record the time required for the ball to fall at constant velocity through a specified distance between reference marks. ● Use the Stoke's law for the sphere falling in a fluid under effect of gravity, to estimate the viscosity of the liquid.		
3	Surface tension of liquid - Capillary rise method	● Clamp a clean capillary tube by dipping its lower end into the liquid in the beaker. ● Measure the rise of water in the tube using a traveling microscope. ● Also measure the radius of the capillary tube using the traveling microscope and estimate the surface tension of the liquid. ● Density of the liquid can be determined using Hare's apparatus of can be given		
4	Density of the liquid using manometer	● Fill a manometer tube partially with water. Pour the given oil (or any liquid which does not mix with water) into the left arm of the tube until the oil–water interface is at the midpoint. Both arms of the tube are open to the air. ● Measure the heights of the oil and water using a traveling microscope and hence estimate the density of the oil assuming that of water. ● Example 12.4 of book 1		
5	Verification of Boyle's law and Charle's law			

		- Boyle's law ($PV = \text{a constant}$) states that at a constant temperature, volume of a gas is inversely proportional to pressure. - Determine the volume - pressure relation at constant temperature using the water column. - Plot the pressure versus volume graph and verify Boyle's law. - Verify the law at minimum two different temperatures. - Charles's law ($V/T = \text{a constant}$) states that at constant pressure, volume is directly proportional to temperature. - In this experiment determine the temperature - volume relation at constant pressure using the water column. - Plot the temperature versus volume graph and verify the Charles's law. - Verify the law at minimum two different pressures.		
	6	Verification of Gay-Lussac's law - Gay-Lussac's law ($P/T = \text{a constant}$) states that at constant volume, pressure is directly proportional to temperature. - In this experiment determine the temperature - pressure relation at constant pressure using metallic bulb and water column or pressure gauge or using Jolly's bulb apparatus. - Plot the temperature versus volume graph and verify the Charles's law.		
	7	Thermal conductivity by Searle's method Determine the thermal conductivity of copper or any other metal using Searle's method / apparatus.		
	8	Temperature coefficient of resistance of a metal - Resistance of metals increases with increase in temperature. - Measure the resistance of the metal coil, using Carey Foster's bridge or Potentiometer or any other suitable method, as a function of temperature from 100 degree Celsius to room temperature.		

		Plot graph and find the temperature coefficient of resistance.		
9	Thermo emf of a Thermocouple	Study the variation of thermo emf of a thermocouple as a function of temperature of the hot junction while maintaining the cold junction at 0 degree Celsius.		
10	Newton's law of cooling	- According to Newton's law of cooling, the rate of heat loss of a hot body is proportional to the difference in temperature between the body and the surroundings. - The calorimeter is filled with hot water and the variation in temperature is noted as a function of time. - Cooling rate graph is plotted and law is verified. - Emissivity of the surface of the calorimeter can also be determined. - ExpEYES with PT1000 sensor may be used to record the temperature. https://expeyes.in/experiments/thermal/cooling.html		
11	Characteristics of NTC thermistor	- Resistance of Negative Temperature Coefficient (NTC) thermistors decreases with increase in temperature. - Measure the resistance of the thermistor, using Carey Foster's bridge or Potentiometer or ExpEYES or any other suitable method, as a function of temperature from 100 degree Celsius to room temperature. - Plot the graph and study the characteristics.		
12	Melting point of wax	- Fill a test tube with wax until half and use a thermometer inside the wax / test tube to measure wax temperature. Avoid the thermometer touching the test tube. - Immerse the test tube in a water bath with the help of a stand, in such a way that the wax is below the water level. - Use a suitable flame / heating rate and measure the wax temperature as a function of time at a suitable time interval. - Plot temperature versus time graph. ExpEYES and PT1000 sensor may be used to record the temperature. https://expeyes.in/experiments/thermal/cooling.html		

		The temperature increases initially and remains constant until the wax melts completely. The flat temperature gives the melting point of wax (The melting point depends on the type of wax used)		
13	Young's Modulus of the Material of a Given Bar: Uniform Bending	- Use an optic lever and telescope. Take measurements for a minimum of two lengths. Obtain the elevation (e) from the shift (s) in the telescope reading and calculate Y from it. - For each length of the bar, plot the load-elevation graph (using GeoGebra) and obtain m/e, and then calculate Y from it.		
14	Torsion Pendulum- Determination of the Moment of Inertia and Rigidity Modulus.	- Using identical masses on the disc, determine the moment of inertia of the disc. - Verify the moment of inertia by direct method, $I = \frac{1}{2}MR^2$ - Using I, calculate rigidity modulus of the material of the wire, $n = \frac{8\pi l}{r^4} \frac{L}{T^2}$		
15	Static torsion Rigidity modulus	Using Searle's static torsion apparatus, determine the rigidity modulus of the material of the rod.		

Books and References:

- University Physics with Modern Physics (Edn.15) by Hugh D. Young & Roger A. Freedman (Book 1)
- Heat and Thermodynamics, 7th Edn.- Mark W Zemansky and Richard H Dittman - McGraw-Hill (Book 2)
- Heat and Thermodynamics - D. S. Mathur - S Chand Publishers (Book 3)
- Berkeley Physics Course : Vol.1 : Mechanics, 2ndEdn. – Kittel et al. – McGraw-Hill (Book 4)

Mapping of COs with PSOs and POs :

	PS O1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	0	0	1	1	2	0	0	1	0	0	0
CO 2	2	2	1	0	1	1	2	0	0	1	0	0	0

CO 3	2	1	2	0	2	1	2	0	0	1	1	0	0
CO 4	2	1	2	0	2	1	2	0	0	1	1	0	0
CO 5	2	2	2	0	2	2	2	1	0	1	1	0	0
CO 6	2	2	1	0	2	2	2	0	0	1	1	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory /Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	